Notes on "An Introduction to Manifolds" by Tu

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1 Smooth Functions on Euclidean Space

In an introductory calculus course, we had worked with $\mathbb{R}^n$. $\mathbb{R}^n$ is a manifold, but it also is equipped with a natural coordinate system, unlike many other manifolds. It turns out that a lot of stuff from calculus that was done was only possible because there existed a corresponding global coordinate system. For example, it is impossible to integrate functions on a manifold in general, since integration requires a coordinate system. The objects that can be integrated are differential forms.

The goal is to recast calculus in $\mathbb{R}^n$ in a way independent of coordinates to generalize to manifolds.

1.1 Smooth vs Analytic Functions

Write coordinates on $\mathbb{R}^n$ as $x^1, \cdots, x^n$ and let $p = (p^1, \cdots, p^n)$ be a point in an open set of $U \subset \mathbb{R}^n$. Index using indices as superscripts (there is reason for this).

Definition

Let $k \in \mathbb{Z}_0^+$. A real-valued function $f : U \to \mathbb{R}^n$ is said to be C^k at $p \in U$ if its partial derivatives

$$\frac{\partial^j f}{\partial x^{i_1} \cdots \partial x^{i_j}}$$

of all orders $j \leq k$ exists and are continuous at p. The function $f : U \to \mathbb{R}^n$ is C^∞ at $p \in U$ if it is C^k for all $k \geq 0$. A vector-valued function $f : U \to \mathbb{R}^m$ is said to be C^k if all of its component functions $f^1, \cdots, f^m$ are C^k at p. We say that $f : U \to \mathbb{R}^m$ is C^k on U if it is C^k for every point $p \in U$. The definition for C^∞ for vector-valued functions are defined in a similar manner. C^∞ and smooth are equivalent.

A neighborhood of a point in $\mathbb{R}^n$ is an open set containing the point. The function f is real analytic at p if in some neighborhood of p it is equal to its

Taylor series at p, i.e.

$$f(x) = f(p) + \sum_i \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \frac{1}{2!}\sum_{i,j} \frac{\partial^2 f}{\partial x^i \partial x^j}(p)(x^i - p^i)(x^j - p^j)$$
$$+ \cdots + \frac{1}{k!}\sum_{i_1,\cdots,i_k} \frac{\partial^k f}{\partial x^{i_1}\cdots\partial x^{i_j}}(p)(x^{i_1} - p^{i_1})\cdots(x^{i_k} - p^{i_k}) + \cdots$$

All real-analytic functions are C^∞, but not all C^∞ functions are real-analytic. Consider the function

$$f(x) = \begin{cases} e^{-1/x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

It is relatively easy to show via induction that $f(x)$ is C^∞. However, note that at $x = 0$, $f(0) = 0$ and $f^{(k)}(0) = 0$ and so its Taylor series expansion should be 0. However, for $x > 0$, $f(x) = e^{-1/x} \neq 0$ and so there does not exist a neighborhood around $x = 0$ such that $f(x)$ is identical to its Taylor series.

1.2 Taylor's Theorem with Remainders

While a C^∞ is not necessarily equal to its Taylor series, there is a Taylor's theorem with remainder that is good enough. In the theorem below, the Taylor series will only consist of the constant term $f(p)$

Taylor's theorem with remainders

Let f be C^∞ function on an open convex subset $U \subset \mathbb{R}^n$. Let $p = (p^1, \cdots, p^n) \in U$. Then there exists functions $g_1(x), \cdots, g_n(x) \in C^\infty(U)$ such that

$$f(x) = f(p) + \sum_{i=1}^n (x^i - p^i) g_i(x)$$

where

$$g_i(p) = \frac{\partial f}{\partial x^i}(p)$$

Proof: For any point $x \in U$, there exists a line segment entirely in U which connects x to p, given by $p + t(x - p)$, for $0 \leq t \leq 1$. Thus, $f(p + t(x - p))$ exists and is C^∞ for all points along the line segment. Taking the derivative of this function with respect to t, we get

$$\frac{d}{dt} f(p + t(x - p)) = \sum_{i=1}^n (x^i - p^i) \frac{\partial f}{\partial x^i}(p + t(x - p))$$

Integrating with respect to t from 0 to 1, we get

$$\int_0^1 \frac{d}{dt} f(p + t(x - p)) dt = \int_0^1 \sum_{i=1}^n (x^i - p^i) \frac{\partial f}{\partial x^i}(p + t(x - p)) dt$$

The LHS evaluates to

$$\int_0^1 \frac{d}{dt} f(p + t(x-p))dt = f(p + t(x-p))\Big|_0^1$$

and the RHS evaluates to

$$\int_0^1 \sum_{i=1}^n (x^i - p^i) \frac{\partial f}{\partial x^i}(p + t(x-p))dt = \sum_{i=1}^n (x^i - p^i) \int_0^1 \frac{\partial f}{\partial x^i}(p + t(x-p))dt$$

If we define $g_i(x)$ to be

$$g_i(x) = \int_0^1 \frac{\partial f}{\partial x^i}(p + t(x-p))dt$$

we see that $g_i(x)$ is C^∞ as f is also C^∞. Combining the results, we have that

$$f(x) - f(p) = \sum_{i=1}^n (x^i - p^i) g_i(x)$$

Now finally, notice that when $x = p$, $p + t(x-p) = p + t(p-p) = p$ and so

$$g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p) \int_0^1 dt = \frac{\partial f}{\partial x^i}(p)$$

Exercise 1.2

Let $f(x)$ be

$$f(x) = \begin{cases} e^{-1/x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

Show by induction that for $x > 0$ and $k \geq 0$, the kth derivative $f^{(k)}(x)$ is of the form $p_{2k}(1/x)e^{-1/x}$ for some polynomial $p_{2k}(y)$ of degree $2k$ in y. Prove that f is C^∞ on $\mathbb{R}$ and that $f^{(k)}(0) = 0$ for all $k \geq 0$

Clearly, for $x > 0$, $f(x) = e^{-1/x}$. In the base case $n = 0$, $f(x) = e^{-1/x}$ and so $p_0 = 1$ is a polynomial of degree 0. Now suppose that for some $n = k$, $f^{(k)}(x) = p_{2k}(1/x)e^{-1/x}$ for some polynomial $p_{2k}(y)$ of degree $2k$ in y. Then

$$\begin{aligned} f^{(k+1)}(x) &= \left(p'_{2k}(1/x) \cdot -\frac{1}{x^2} e^{-1/x}\right) + \left(p_{2k}(1/x)e^{-1/x} \cdot \frac{1}{x^2}\right) \\ &= e^{-1/x} (1/x)^2 \left(p_{2k}(1/x) - p'_{2k}(1/x)\right) \end{aligned}$$

Letting $y = 1/x$, we see that

$$\begin{aligned} e^{-1/x} (1/x)^2 \left(p_{2k}(1/x) - p'_{2k}(1/x)\right) &= e^{-1/x} y^2 \left(p_{2k}(y) - p'_{2k}(y)\right) \\ &= e^{-1/x} p_{2k+2}(y) = e^{-1/x} p_{2(k+1)}(1/x) \end{aligned}$$

Self-evidently, when $x < 0$, $f(x) = 0$ is C^∞ and likewise, when $x > 0$, $f(x) = e^{-1/x}$ is C^∞. The only contention arises when dealing with $x = 0$, as one must show all the derivatives of $f(x)$ is continuous there.

To show that $f(x)$ is C^∞ at 0, we need to show that for $k \geq 0$

$$\lim_{x \to 0} f^{(k)}(x) = 0$$

Clearly, since $f(x) = 0$ for $x < 0$, we have that

$$\lim_{x \to 0^-} f^{(k)}(x) = 0$$

Since $f(x) = e^{-1/x}$, $f^{(k)}(x) = p_{2k}(1/x)e^{-1/x}$ for some polynomial $p_{2k}(y)$ of degree y.

$$\lim_{x \to 0^+} f^{(k)}(x) = \lim_{y \to \infty} p_{2k}(y)e^{-y} = 0$$

Therefore, $f(x)$ is C^∞ at 0.

Exercise 1.6

Prove that if $f : \mathbb{R}^2 \to \mathbb{R}$ is C^∞, then there exists functions g_{11}, g_{12}, g_{22} such that

$$\begin{aligned} f(x,y) = f(0,0) + \frac{\partial f}{\partial x}(0,0)x + \frac{\partial f}{\partial y}(0,0)y + x^2 g_{11}(x,y) \\ + xy g_{12}(x,y) + y^2 g_{22}(x,y) \end{aligned}$$

Since we know that f is C^∞, we know from Taylor's theorem with Remainder that there exists C^∞ functions g_1, g_2 such that

$$\begin{aligned} f(x,y) &= f(0,0) + (x-0)g_1(x,y) + (y-0)g_2(x,y) \\ &= f(0,0) + xg_1(x,y) + yg_2(x,y) \end{aligned}$$

and

$$\begin{aligned} g_1(0,0) &= \frac{\partial f}{\partial x}(0,0) \\ g_2(0,0) &= \frac{\partial f}{\partial y}(0,0) \end{aligned}$$

Applying Taylor's theorem with remainder to the function $g_1(x,y)$, we see that there exists functions h_{11}, h_{12} in C^∞ such that

$$\begin{aligned} g_1(x,y) &= g_1(0,0) + (x-0)h_{11}(x,y) + (y-0)h_{12}(x,y) \\ &= \frac{\partial f}{\partial x}(0,0) + xh_{11}(x,y) + yh_{12}(x,y) \end{aligned}$$

Similarly, applying it to $g_2(x,y)$, we see that there exists functions h_{21}, h_{22} in C^∞ such that

$$\begin{aligned} g_2(x,y) &= g_2(0,0) + (x-0)h_{21}(x,y) + (y-0)h_{22}(x,y) \\ &= \frac{\partial f}{\partial y}(0,0) + xh_{21}(x,y) + yh_{22}(x,y) \end{aligned}$$

Plugging these results in, we have that

$$\begin{aligned} f(x,y) = f(0,0) &+ x\left(\frac{\partial f}{\partial x}(0,0) + xh_{11}(x,y) + yh_{12}(x,y)\right) \\ &+ y\left(\frac{\partial f}{\partial y}(0,0) + xh_{21}(x,y) + yh_{22}(x,y)\right) \\ f(x,y) = f(0,0) &+ \frac{\partial f}{\partial x}(0,0)x + \frac{\partial f}{\partial y}(0,0)y + x^2 h_{11}(x,y) \\ &+ xy\left(h_{12}(x,y) + h_{21}(x,y)\right) + y^2 h_{22}(x,y) \end{aligned}$$

Now let $g_{11}(x,y) = h_{11}(x,y)$, $g_{12}(x,y) = h_{12}(x,y) + h_{21}(x,y)$, and $g_{22}(x,y) = h_{22}(x,y)$. This completes the proof.

Exercise 1.7

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a C^∞ function with

$$f(0,0) = \frac{\partial f}{\partial x}(0,0) = \frac{\partial f}{\partial y}(0,0) = 0$$

Define a function $g : \mathbb{R}^2 \to \mathbb{R}$ by

$$g(t,u) = \begin{cases} \dfrac{f(t,tu)}{t}, & t \neq 0 \\ 0, & t = 0 \end{cases}$$

Prove that $g(t,u)$ is C^∞ for $(t,u) \in \mathbb{R}^2$

Recall from the earlier exercise 1.6 that since f is C^∞, then there exists functions g_{11}, g_{12}, g_{22} such that

$$\begin{aligned} f(x,y) = f(0,0) &+ \frac{\partial f}{\partial x}(0,0)x + \frac{\partial f}{\partial y}(0,0)y + x^2 g_{11}(x,y) \\ &+ xyg_{12}(x,y) + y^2 g_{22}(x,y) \end{aligned}$$

Plugging in the earlier conditions, we have that

$$f(x,y) = x^2 g_{11}(x,y) + xyg_{12}(x,y) + y^2 g_{22}(x,y)$$

For $x = t, y = tu$, we have that

$$f(t,tu) = t^2 g_{11}(t,tu) + t^2 u g_{12}(t,tu) + (tu)^2 g_{22}(t,tu)$$

and so finally, we have that

$$\begin{aligned}\frac{f(t,tu)}{t} &= tg_{11}(t,tu) + tug_{12}(t,tu) + tu^2 g_{22}(t,tu) \\ &= t\left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right)\end{aligned}$$

Recall the earlier definition of $g(t,u)$. The derivative of this function with respect to t is

$$\begin{aligned}\frac{\partial g}{\partial t}(t,u) &= \left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right) \\ +t\left(\frac{\partial g_{11}}{\partial x}(t,tu)\frac{\partial x}{\partial t} + \frac{\partial g_{11}}{\partial y}(t,tu)\frac{\partial y}{\partial t} + u\left(\frac{\partial g_{12}}{\partial x}(t,tu)\frac{\partial x}{\partial t} + \frac{\partial g_{12}}{\partial y}(t,tu)\frac{\partial y}{\partial t}\right)\right) \\ +tu^2\left(\frac{\partial g_{22}}{\partial x}(t,tu)\frac{\partial x}{\partial t} + \frac{\partial g_{22}}{\partial y}(t,tu)\frac{\partial y}{\partial t}\right) \\ &= \left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right) \\ +t\left(\frac{\partial g_{11}}{\partial x}(t,tu) + u\frac{\partial g_{11}}{\partial y}(t,tu) + u\left(\frac{\partial g_{12}}{\partial x}(t,tu) + u\frac{\partial g_{12}}{\partial y}(t,tu)\right)\right) \\ +tu^2\left(\frac{\partial g_{22}}{\partial x}(t,tu) + u\frac{\partial g_{22}}{\partial y}(t,tu)\right)\end{aligned}$$

Similarly, the derivative of g with respect to u is

$$\begin{aligned}\frac{\partial g}{\partial u}(t,u) &= t\left(\frac{\partial g_{11}}{\partial x}(t,tu)\frac{\partial x}{\partial u} + \frac{\partial g_{11}}{\partial y}(t,tu)\frac{\partial y}{\partial u} + g_{12}(t,tu) + u\left(\frac{\partial g_{12}}{\partial x}(t,tu)\frac{\partial x}{\partial u} + \frac{\partial g_{12}}{\partial y}(t,tu)\frac{\partial y}{\partial u}\right)\right) \\ &\quad +t\left(2ug_{22}(t,tu) + u^2\left(\frac{\partial g_{22}}{\partial x}(t,tu)\frac{\partial x}{\partial u} + \frac{\partial g_{22}}{\partial y}(t,tu)\frac{\partial y}{\partial u}\right)\right) \\ &= t\left(t\frac{\partial g_{11}}{\partial y}(t,tu) + g_{12}(t,tu) + u\left(t\frac{\partial g_{12}}{\partial y}(t,tu)\right) + 2ug_{22}(t,tu) + u^2\left(t\frac{\partial g_{22}}{\partial y}(t,tu)\right)\right)\end{aligned}$$

Notice where $t \neq 0$

$$g(t,u) = \frac{f(t,tu)}{t} = t\left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right)$$

is C^∞ for all points (t,u) where $t \neq 0$ since all the component parts are also C^∞. It is also continuous at $t = 0$ since

$$\lim_{t\to 0} t\left(g_{11}(t,tu) + ug_{12}(t,tu) + u^2 g_{22}(t,tu)\right) = 0$$

since g_{11}, g_{12}, g_{22} are C^∞ and so must converge to a definite value. Finally, notice that the partial derivatives

$$\nabla g(t,u) = \left(\frac{\partial g}{\partial t}(t,u), \frac{\partial g}{\partial u}(t,u)\right)$$

are composed of functions that at the limit is C^∞ at $t = 0$ and so the partial derivatives are C^∞ at $t = 0$.

Tangent vectors in $\mathbb{R}^n$ as Derivations

We typically express a vector v at a point $p \in \mathbb{R}^n$ as a column of numbers

$$v = \begin{bmatrix} v^1 \\ \vdots \\ v^n \end{bmatrix}$$

or as a row of numbers bounded by angular brackets

$$v = \langle v^1, \cdots, v^n \rangle$$

Recall that a secant plane to a surface in $\mathbb{R}^3$ is given by three non-collinear points on a surface, and as the three points approach a point p on the surface, the plane at the limit position of the secant plane is called the tangent plane at p. Intuitively, we can understand this as the plane simply "touching" the surface at this point. A vector at p is tangent to the surface if it lies in the tangent plane at p.

Directional Derivative

In $\mathbb{R}^n$, we can visualize the tangent space $T_p(\mathbb{R}^n)$ at p as the vector space of all vectors emanating from p. Since we can express the vector space with a list of n numbers, it is identical to $\mathbb{R}^n$. To distinguish between points and vectors then, we let a point $p \in \mathbb{R}^n$ be written as $p = \left(p^1, \cdots, p^n\right)$ and a vector $v \in T_p(\mathbb{R}^n)$ be written as

$$v = \begin{bmatrix} v^1 \\ \vdots \\ v^n \end{bmatrix} \quad \text{or} \quad v = \langle v^1, \cdots, v^n \rangle$$

Typically, the standard basis for $\mathbb{R}^n$ or $T_p(\mathbb{R}^n)$ is denoted by $e_1, \cdots, e_n$. Then, we can express $v \in T_p(\mathbb{R}^n)$ as $v = \sum v^i e_i$ for some $v^i \in \mathbb{R}$. Elements of $T_p(\mathbb{R}^n)$ are called tangent vectors at p in $\mathbb{R}^n$.

The line through a point $p = \left(p^1, \cdots, p^n\right)$ with a direction $v = \langle v^1, \cdots, v^n \rangle$ has parametrization

$$c(t) = \left(p^1 + tv^1, \cdots, p^n + tv^n\right)$$

If f is C^∞ at p and v is a tangent vector at p, then the directional derivative of f in the direction of v at p is defined to be

$$D_v f = \lim_{t \to 0} \frac{f(c(t)) - f(p)}{t} = \left.\frac{d}{dt}\right|_{t=0} f(c(t))$$

By the chain rule,

$$D_v f = \sum_{i=1}^{n} \frac{dc^i}{dt}(0) \frac{\partial f}{\partial x^i}(p)$$

We know that $c'(t) = v$ and so

$$D_v f = \sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}(p) = v \cdot \nabla f(p)$$

The reason the notation $D_v f$ is used, instead of $D_v f(p)$, is that the point p is already encoded in the vector v, since $v \in T_p\mathbb{R}^n$. v is anchored at p. Thus, the directional derivative is inherently taken at p. Moreover, $D_v f$ is understood as a number, the directional derivative at p, and not a function of p.

We write

$$D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\bigg|_p$$

to be the map that sends a function f to the number $D_v f$. The association $v \mapsto D_v$ of the directional derivative D_v to the tangent vector v gives a way to characterize tangent vectors as operators on functions.

Germs of Functions

Equivalence Relation

A relation on a set S is a subset $R \subset S \times S$. R is an equivalence relation if it satisfies for all $x, y, z \in S$

1. (Reflexivity) $x \sim x$
2. (Symmetry) $x \sim y$ if and only if $y \sim x$
3. (Transitivity) $x \sim y$ and $y \sim z$ implies $x \sim z$

If two functions agree on some neighborhood of a point p, they would have the same directional derivatives at p, suggesting that an equivalence relation can be defined on the C^∞ functions defined on some neighborhood of p.

Germs of Functions

Consider the set of all pairs (f, U), where U is a neighborhood of p and $f : U \to \mathbb{R}$ is C^∞. We say that two pairs (f, U) and (g, V) are equivalent to each other if there exists an open set $W \subset U \cap V$ containing p such that $f = g$ when restricted to W. The equivalence class of (f, U) is called the germ of f at p. We write $C_p^\infty(\mathbb{R}^n)$ (or C_p^∞ if there is no possibility of confusion) for the set of all germs of C^∞ functions on $\mathbb{R}^n$ at p.

Consider the function

$$f(x) = \frac{1}{1-x}$$

over the domain $\mathbb{R} - \{1\}$ and

$$g(x) = 1 + x + x^2 + x^3 + \cdots$$

with the domain $(-1, 1)$. f and g have the same germ at any point $p \in (-1, 1)$.

Algebra

An algebra over a field K is a vector space A over K with the multiplication map

$$\mu : A \times A \to A$$

denoted typically by $\mu(a, b) = a \cdot b$, such that for all $a, b, c \in A$ and $r \in K$

1. (Associativity) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. (Distribution) $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + a \cdot b$
3. (Homegeneuity) $r(a \cdot b) = (ra) \cdot b = a \cdot (rb)$

This is equivalent to stating that an algebra over a field K is a ring A that is also a vector space over K such that the ring multiplication satisfies the homogeneity condition.

Linear Operator

A map $L : V \to W$ between vector spaces over a field K is called a linear operator if for any $r \in K$ and $u, v \in V$

1. $L(u + v) = L(u) + L(v)$
2. $L(rv) = rL(v)$

If A and A' are algebras over a field K, then an algebra homomorphism is a linear map $L : A \to A'$ that preserves the algebra multiplication, i.e. $L(a \cdot b) = L(a) \cdot L(b)$ for all $a, b \in A$.

The addition and multiplication of functions induces the corresponding operations on C_p^∞, making it into an algebra over $\mathbb{R}$.

Derivation at a Point

For each tangent vector v at a point $p \in \mathbb{R}^n$, the directional derivative at p gives a map of real vector spaces

$$D_v : C_p^\infty \to \mathbb{R}$$

Consider two functions $f, g \in C_p^\infty$ and a scalar $r \in \mathbb{R}$. Recall that since

$$D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\bigg|_p$$

we have that

$$\begin{aligned} D_v(f + g) = \sum_{i=1}^{n} v^i \frac{\partial (f + g)}{\partial x^i}(p) &= \sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}(p) + \sum_{i=1}^{n} v^i \frac{\partial g}{\partial x^i}(p) \\ &= D_v f + D_v g \end{aligned}$$

Similarly,

$$D_v(rf) = \sum_{i=1}^{n} v^i \frac{\partial(rf)}{\partial x^i}(p) = r\sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}(p) = rD_v f$$

Therefore, D_v is a linear map (D_v is $\mathbb{R}$-linear). Furthermore, we know that D_v satisfies the Leibniz rule

$$D_v(fg) = (D_v f)g(p) + f(p)D_v g$$

since

$$D_v(fg) = \sum_{i=1}^{n} v^i \frac{\partial(fg)}{\partial x^i}(p) = \sum_{i=1}^{n} v^i \left(\frac{\partial f}{\partial x^i}(p)g(p) + f(p)\frac{\partial g}{\partial x^i}(p)\right)$$
$$= g(p)\sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}(p) + f(p)\sum_{i=1}^{n} v^i \frac{\partial g}{\partial x^i}(p) = (D_v f)g(p) + f(p)(D_v g)$$

Any linear map $D : C_p^\infty \to \mathbb{R}$ that satisfies the Leibniz rule is referred to as a *derivation at* p or a *point-derivation* of C_p^∞. Denote the set of all derivations at p by $\mathcal{D}_p(\mathbb{R}^n)$. This space is a vector space.

We know that all directional derivatives at p are all derivations at p, so there exists a map

$$\phi : T_p(\mathbb{R}^n) \to \mathcal{D}_p(\mathbb{R}^n)$$

given by

$$v \mapsto D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\bigg|_p$$

We can deduce that ϕ is a linear map since for vectors $v, w \in T_p(\mathbb{R}^n)$ and $r \in \mathbb{R}$

$$\phi(v+w) = D_{v+w} = \sum_{i=1}^{n} (v+w)^i \frac{\partial}{\partial x^i}\bigg|_p = \sum_{i=1}^{n} (v^i + w^i)\frac{\partial}{\partial x^i}\bigg|_p$$
$$= \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\bigg|_p + \sum_{i=1}^{n} w^i \frac{\partial}{\partial x^i}\bigg|_p = D_v + D_w = \phi(v) + \phi(w)$$

and

$$\phi(rw) = D_{rw} = \sum_{i=1}^{n} (rw)^i \frac{\partial}{\partial x^i}\bigg|_p = \sum_{i=1}^{n} r(w^i)\frac{\partial}{\partial x^i}\bigg|_p$$
$$= r\sum_{i=1}^{n} w^i \frac{\partial}{\partial x^i}\bigg|_p = rD_w = r\phi(w)$$

Lemma 2.1

If D is a point-derivation of C_p^∞, then $D(c) = 0$ for any constant function c.
Proof: Since every point-derivation is $\mathbb{R}$-linear, we know that $D(c) = cD(1)$, and so what we need to show is that $D(1) = 0$. We know by Leibniz rule that

$$D(1) = D(1 \cdot 1) = D(1) \cdot 1 + 1 \cdot D(1) = 2D(1)$$

and so $D(1) = 0$. Thus, $D(c) = 0$.

The Kronecker delta δ is a convenient notation used defined in the following manner

$$\delta_i^j = \begin{cases} 1 & i = j, \\ 0 & i \neq j \end{cases}$$

Theorem 2.2

The linear map $\phi : T_p(\mathbb{R}^n) \to \mathcal{D}_p(\mathbb{R}^n)$ given by $v \mapsto D_v$ is an isomorphism of vector spaces.
Proof: To show that this is an isomorphism, we have to show that it is bijective and that it is a linear map. We have already shown it is a linear operator, so now, we must show it is bijective.

To show that it is injective, we need to show that the kernel of ϕ is solely the zero vector. Let $x^j : \mathbb{R}^n \to \mathbb{R}$ be the coordinate function sending a point q to its coordinate q^j, i.e. $x^j(q) = q^j$ for $q \in \mathbb{R}$. Let $D_v = 0$. Applying this operator to the coordinate function, we have that

$$D_v x^j = \sum_{i=1}^n v^i \frac{\partial}{\partial x^i}\bigg|_p x^j = \sum_{i=1}^n v^i \delta_i^j = v^j = 0$$

Therefore, $v = 0$ and so ϕ is injective.

To show that it is surjective, let D be a derivation at p, i.e. $D \in \mathcal{D}_p(\mathbb{R}^n)$ and let (f, V) be a representative of a germ in C_p^∞. By making V as small as necessary, we can assume that V is an open ball. By Taylor's Theorem with remainder, we know there exists C^∞ functions $g_1(x), \cdot, g_n(x)$ in a neighborhood of p such that

$$f(x) = f(p) + \sum_{i=1}^n (x^i - p^i) g_i(x), \quad g_i(p) = \frac{\partial f}{\partial x^i}(p)$$

Apply D to both sides

$$\begin{aligned} Df(x) = D\left(f(p) + \sum_{i=1}^n (x^i - p^i) g_i(x)\right) &= Df(p) + D\left(\sum_{i=1}^n (x^i - p^i) g_i(x)\right) \\ &= Df(p) + \sum_{i=1}^n D\left((x^i - p^i) g_i(x)\right) \end{aligned}$$

From Leibniz rule, we know that

$$\begin{aligned} D\left((x^i - p^i)g_i(x)\right) &= D(x^i - p^i)g_i(p) + (p^i - p^i)Dg_i(x) \\ &= (Dx^i)g_i(p) - (Dp^i)g_i(p) \end{aligned}$$

Since $f(p)$ and p^i are constants, $Df(p) = Dp^i = 0$ and so

$$Df(x) = \sum_{i=1}^{n}(Dx^i)g_i(p) = \sum_{i=1}^{n}(Dx^i)\frac{\partial f}{\partial x^i}(p)$$

Thus, there exists a vector $v = \langle Dx^1, \cdots, Dx^n\rangle$ such that $D = D_v$. *Intuitively, a derivation D at a point p is completely determined by how it acts on the coordinate functions $x^1, \cdots, x^n$. The values Dx^i give the components of the vector v, showing that D is just the directional derivative in that direction.*

This theorem allows us to identify the tangent vectors at p with the derivations at p. Under the isomorphism $T_p(\mathbb{R}^n) \cong \mathcal{D}_p(\mathbb{R}^n)$, the standard basis $e_1, \cdots, e_n$ for $T_p(\mathbb{R}^n)$ corresponds to the set $\partial/\partial x^1|_p, \cdots, \partial/\partial x^n|_p$ of partial derivatives. Thus, we can write a tangent vector $v = \langle v^1, \cdots, v^n\rangle = \sum v^i e_i$ as

$$v = \sum v^i \frac{\partial}{\partial x^i}\bigg|_p$$

While $T_p(\mathbb{R}^n)$ is isomorphic to $\mathcal{D}_p(\mathbb{R}^n)$, there is a reason the latter is preferred over the former, which is that a vector $v \in T_p(\mathbb{R}^n)$ depends on a choice of coordinates to describe it, while a derivation $D \in \mathcal{D}_p(\mathbb{R}^n)$ is independent of coordinates and one solely relies on how it impacts functions to determine its behavior.

Derivation at a Point